AI Agent as defined by IBM, “is a system that autonomously performs tasks by designing workflows with available tools” that stands as one of the great innovations and improvements of current AI development. AI ‘Agents’ derive function from their intuitive and systematic approach in dealing with queries and requests from the user. As previous monolithic models in AI bottlenecked their capabilities of understanding on the limited amount of data they’ve been trained on. AI Agents unlocked access to deeper agentic reasoning using Large Language Models, giving it the capability to form a plan, execute, and reassess all with the cohesive intent to best solve the request. AI Agents derive their speciality in this form of control logic and the implementation of tool-calling, to externally search another database, perform mathematical calculations, and even properly compute and debug code.

As for the types of AI Agents, they vary mainly by both function and decisional logic. At the most basic forms of Agents are most ‘Simple Reflex Agents’, which act simply as executions to be triggered. They act solely on predefined conditions without the need to learn or memorize information. Best used for simple automation in which presets will be highly controlled and predictable, allowing for immediate response and action. The more intricate the Agent becomes, becoming adaptative, methodizing and goal planning, and even intercommunication between other Agents; determine the type of AI Agent and clarify which model is best used for specific applications.

Real-world applications of AI are far and few between as the whole modern world begins to embark on the AI revolution. From simple automation using ‘Simple Reflex Agents’ to more esoteric language models, AI acts as a shining beacon, drawing in waves of consumers and creators alike. My group’s AI Agent looks to solve a niche in the entrepreneurial business market as a ‘Sales Data & Marketing Advisor’. The struggle in dealing with small or local business comes in the scope in which you perceive options and ideas; and primarily in the scope of E-commerce business they often lack the resources or expertise to analyze sales data and develop strategic business plans. Manual analysis is time-consuming, error-prone, and reactive rather than proactive, establishing the niche need of an intelligent system that can interpret and generate actionable fiscal strategies in the name of profitability efficiency. By having a Utility-Based Agent that allows the agent to assess the multitude of variables that

comes with formalizing business plans and curated strategy; we can bridge-the-gap of experience with intelligence and wisdom.

The best tools that came within my research were GPT-4 as the main form of reasoning and Hugging Face for specialized tasks with more conformability. Fast API + SQLAlchemy to ingest and analyze sales data from online E-Commerce storefronts. The goal of the AI agent will first be accessibility with any storefront to ingest real-time sales data, analyze performance trends, and generate fiscal strategies. It will identify product velocity, seasonal shifts/trends, and even customer behavior; then recommend actions like bundling, markdowns, or restocking. Forecasting future sales and encouraging user chatting and questioning are also key goals during the development cycle and integration. As it comes to testing and researching the clearest signs of success would be determined in its capability to pull local data from the storefront and then to call an external search on the best recommendations that align with business outcomes. A good way of testing would be to simulate its given recommendations in a controlled and simulated market to evaluate the reasoning skills of the agent. In essence, it should be able to accurately analyze trends in sales and formulate deep reasoning on whether an accurate prediction could be made based on the data. In all code efficiency reigns supreme, so flattening out the kinks and issues that come with tuning the responses will be important. As for the goal of conversability; a main criteria of success is on the personally curated prompts and reports the Agent can deliver to the user that promotes interaction with the Agent and breathes new perspectives onto the user. In terms of scalability success; its important the Agent can deal with heavy traffic when dealing with E-commerce websites with sudden bursts of inputs and orders, and to make sure the software can scale up and down as needed for the vastly varying data sets different businesses have.

My research into AI agents, their types, applications, and supporting tools directly shaped the design and functionality of my AI Business Strategy Agent for storefronts. After researching AI agents I realized that the most important tool in any software project is choosing the right agent for the job, but also choosing the right job for the agent. At first my software project was focused on creating a Point of

Sale software, but after researching I realized my main focus should simply be the AI integration for ease of use and real functionality for users.